Computational Physics 2 Classes No.5 - Report

Time-dependent Schroedinger equation (finite difference approach)

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1 Introduction

In this project, the time-dependent Schrödinger equation in two dimensions is solved numerically using a finite difference approach. The system under consideration is a slightly anisotropic harmonic oscillator with $\omega_x = 1$ and $\omega_y = 1.001$.

First, the lowest-energy eigenstates of the Hamiltonian are obtained using an iterative imaginary-time method. These states are then used to construct an initial wave packet, which is propagated in time using the Crank–Nicolson scheme.

The goal of the simulation is to analyze the time evolution of the wave packet, in particular its norm, energy, and expectation values of position, and to study the resulting motion in relation to the oscillator parameters.

1.1 Numerical Parameters

The system was discretized on a two-dimensional square grid of size $N \times N$ with $N = 65$. The spatial domain was chosen as $L = 8$, corresponding to the interval $x, y \in [-4, 4]$. The grid spacing was therefore

$$\Delta x = \Delta y = \frac{L}{N-1} = \frac{8}{64} = 0.125.$$

Dirichlet boundary conditions ($\Psi = 0$) were imposed at the edges of the domain.

Time evolution was performed using the Crank–Nicolson scheme with time step $\Delta t = 0.001$ and 5 iterations per time step. The total simulated time was $t_{\max} = 4\pi$, corresponding to two nominal periods of motion.

All calculations were carried out in atomic units ($\hbar = m = 1$).

2 Results

2.1 Eigenstates and Energies

The first step of the calculation was the determination of the three lowest-energy eigenstates of the Hamiltonian using an iterative imaginary-time method. The obtained energies are:

- Ground state: $E_0 \approx 0.9995$
- First excited state: $E_1 \approx 1.9976$
- Second excited state: $E_2 \approx 1.9986$

These values are in good agreement with the expected analytical results for a two-dimensional harmonic oscillator:

$$E_{n_x,n_y} = \left(n_x + \frac{1}{2}\right)\omega_x + \left(n_y + \frac{1}{2}\right)\omega_y.$$

The small discrepancies are due to the finite-difference discretization.

2.2 Time Evolution of the Initial Wave Packet

The initial wave packet was constructed as:

$$\Psi(0) = \frac{1}{\sqrt{3}}(\Psi_0 + \Psi_1 + i\Psi_2).$$

2.2.1 Norm and Energy Conservation

The norm and average energy as functions of time are shown in Figures 1 and 2.

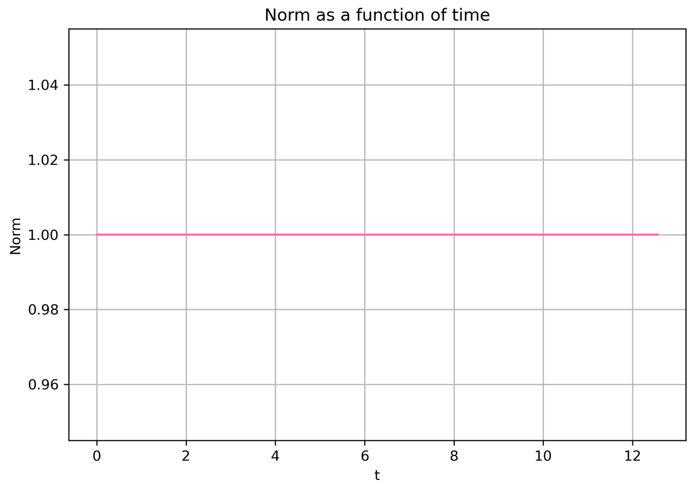

Figure 1: Norm as a function of time.

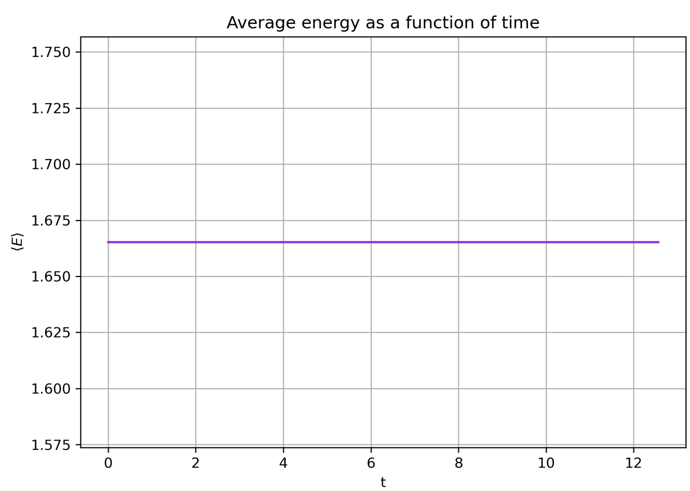

Figure 2: Average energy as a function of time.

The norm remains equal to 1 throughout the simulation, indicating numerical stability of the Crank–Nicolson scheme. The average energy is also conserved, confirming the correctness of the time evolution.

2.2.2 Expectation Values of Position

The expectation values $\langle x \rangle(t)$ and $\langle y \rangle(t)$ are shown in Figure 3.

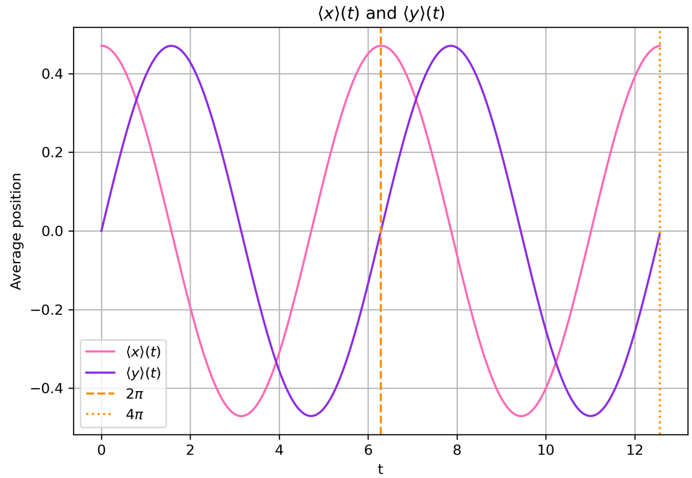

Figure 3: Time dependence of $\langle x \rangle(t)$ and $\langle y \rangle(t)$.

Both quantities exhibit oscillatory behavior with a period close to 2π, as expected for a harmonic oscillator. However, a small phase shift is visible over time.

2.2.3 Trajectory in the $(\langle x \rangle, \langle y \rangle)$ Plane

The trajectory of the wave packet is shown in Figure 4.

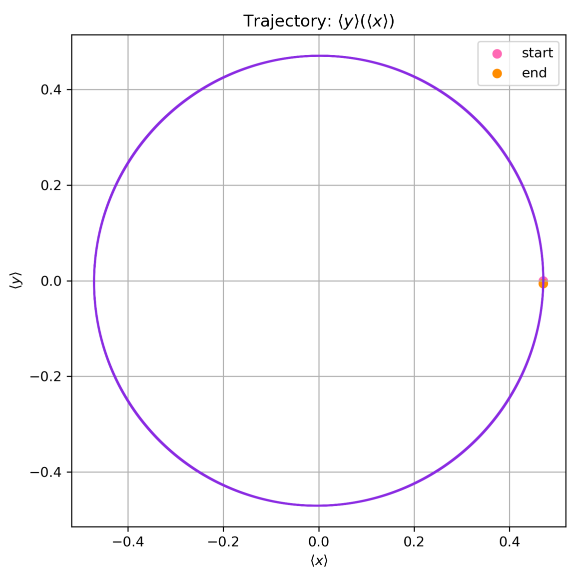

Figure 4: Trajectory $\langle y \rangle(\langle x \rangle)$.

The motion forms an approximately closed loop, corresponding to a rotating wave packet. However, the loop is not perfectly closed. This is due to the slight difference between ω_x and ω_y, which leads to a slow dephasing of the motion.

2.3 Reversed Direction of Rotation

By changing the phase in the initial state, the direction of rotation can be reversed. The comparison is shown in Figure 5.

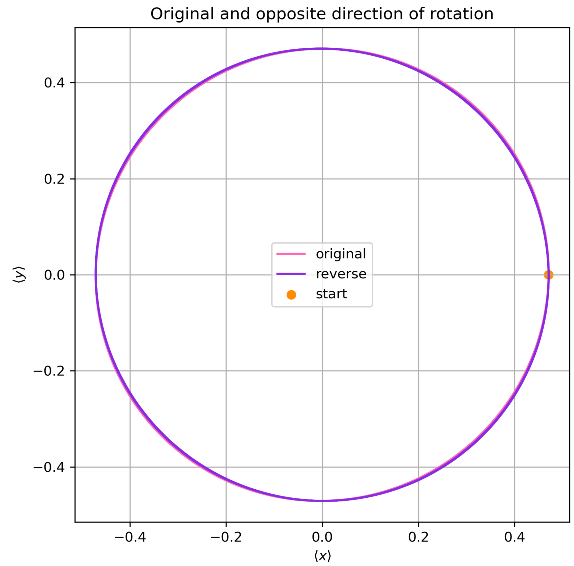

Figure 5: Comparison of original and reversed rotation.

The trajectory is mirrored, confirming that the direction of motion is controlled by the relative phase between the contributing eigenstates.

2.4 Pure Oscillations Along Coordinate Axes

2.4.1 Oscillation in the x Direction

For the initial condition $\Psi(0) \propto \Psi_0 + \Psi_1$, the motion reduces to oscillation along the x axis. Results are shown in Figures 6 and 7.

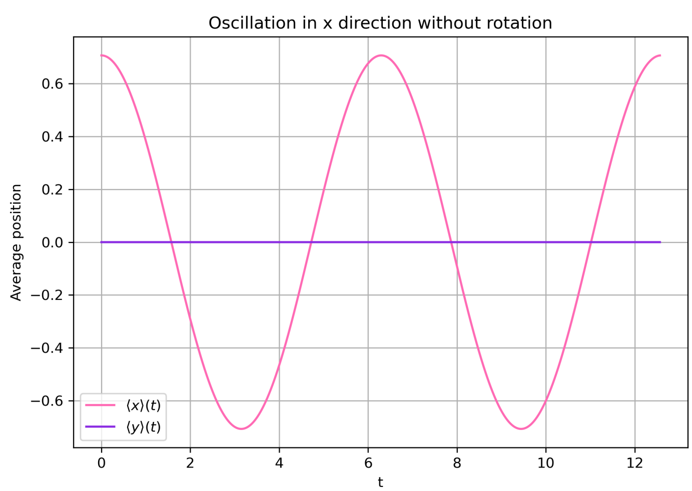

Figure 6: Oscillation in the x direction.

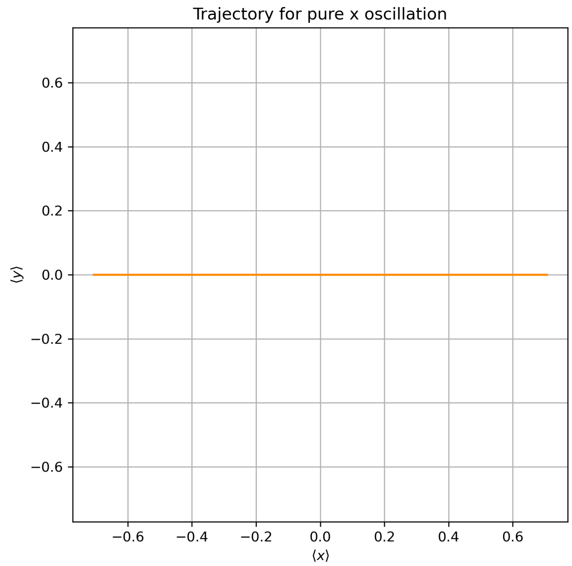

Figure 7: Trajectory for x-only oscillation.

The expectation value $\langle y \rangle$ remains approximately zero.

2.4.2 Oscillation in the y Direction

Similarly, for $\Psi(0) \propto \Psi_0 + \Psi_2$, the motion is confined to the y direction (Figures 8 and 9).

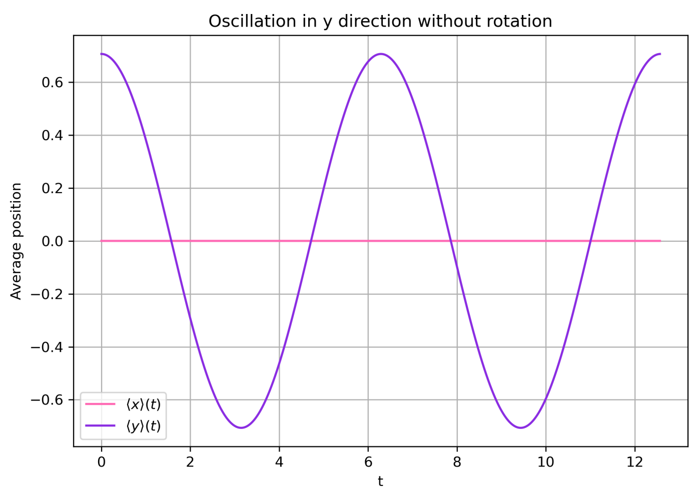

Figure 8: Oscillation in the y direction.

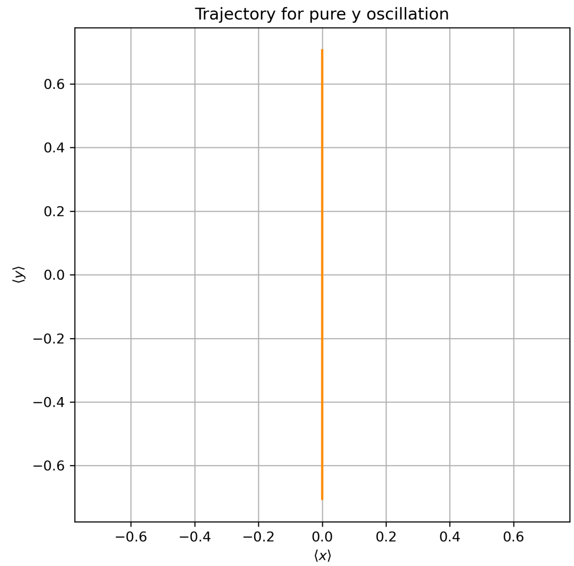

Figure 9: Trajectory for y-only oscillation.

2.5 Evolution of Eigenstates

Finally, the time evolution was studied for initial states equal to individual eigenstates.

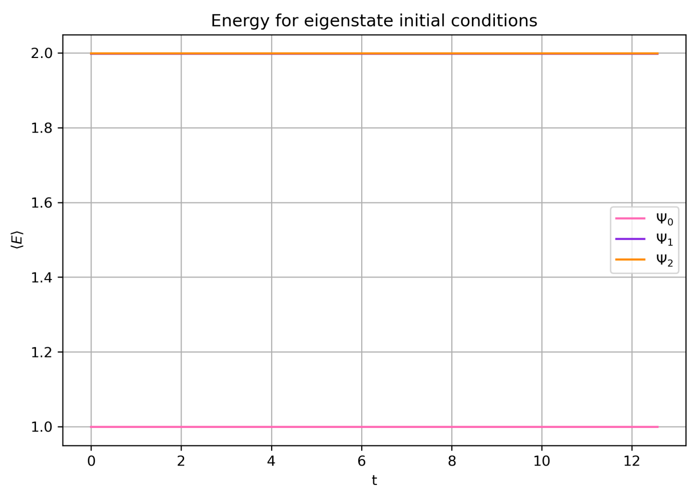

Figure 10: Energy for eigenstate initial conditions.

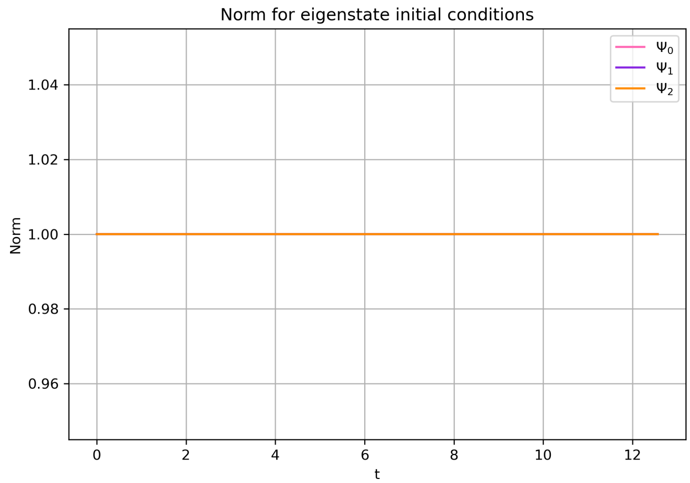

Figure 11: Norm for eigenstate initial conditions.

The energy and norm remain constant in time, as expected. Additionally, the expectation values $\langle x \rangle$ and $\langle y \rangle$ remain approximately zero, indicating that the probability density does not evolve.

This confirms that eigenstates of the Hamiltonian evolve only by a phase factor:

$$\Psi(t) = \Psi_i e^{-iE_i t}.$$

3 Conclusions

In this project, the time-dependent Schrödinger equation for a two-dimensional harmonic oscillator was successfully solved using a finite difference method combined with the Crank–Nicolson time propagation scheme.

The numerical results demonstrate that the method preserves both the norm and the average energy of the system, confirming its stability and accuracy. The constructed wave packet exhibits oscillatory motion with a period close to 2π, as expected. However, due to the slight anisotropy of the potential ($\omega_x \neq \omega_y$), the motion is not perfectly periodic and the trajectory is not exactly closed, showing a slow phase drift.

By modifying the relative phase between eigenstates, it is possible to control the direction of rotation of the wave packet. Additionally, specific superpositions allow for pure oscillations along a single spatial direction without rotational motion.

Finally, simulations with eigenstates as initial conditions confirm that such states remain stationary in terms of probability density, with only a time-dependent phase factor, as predicted by quantum mechanics.

Overall, the numerical approach provides a consistent and physically accurate description of the system dynamics.